

IT8108: Project Design Document & Prototype

Semester:	Semester A, 2026-2027
Tutors:	Haetham Alhaddad (Course Coordinator), Ghassan AlShajjar (Tutor)
Learning Outcomes Covered:	LO1 - Design a user-centric user interface that follows UI and UX best practice. LO2 - Implement complex programs (or Apps) for a mobile platform to a given business requirement. LO3 - Implement best practice, industry standards, professional ethics, diagram designing, programming and documentation conventions during the programming process.
Weighting:	Project Design Document – marked out of 20 and weighted to 10% Project Prototype – marked out of 20 and weighted to 10%
Deadlines:	Project Design Document – 22nd Oct 2026, 11:55 PM Project Prototype – 12th Nov 2026, 11:55 PM
Notes:	Extensions: Requests for extensions should be made 48 hours prior to the deadline and sent to the Course Coordinator. Extensions will only be approved for valid reasons. Extensions are for the whole group. You are only permitted a maximum of one extension per course per semester. Late submissions: All late submissions less than 3 days will be capped at 60%. All late submissions more than 3 days will receive 0 grade.

Project Overview

This project gives you the opportunity to apply the skills learnt in the course by **planning, designing, and developing a real-world mobile application**. In this assessment, you will be engaging in a real software development process that begins with understanding user needs, leading to the creation and demonstration of a functional, well-designed mobile application.

The focus is on delivering a meaningful user experience through thoughtful design, clear functionality, and an organized interface. Each project is based on a real client brief, allowing you to address authentic design challenges while developing collaboration and communication skills within your team.

Following the early stages of the software development life cycle, the project emphasizes user research, feature planning, mockup creation, prototyping, development, and finally a live project demonstration.

Your team will be assessed how effectively you communicate your ideas, justify your design decisions, and translate user needs into an interactive app experience.

Development for this project will target the **iOS platform**, and all applications are expected to be built using **Xcode 26.6 or 27** with compatibility for **iOS 26.0 or later**.

Assessment Description

This is a **group project**; each group should consist of **5 members only**. You will be asked to propose your own team members; however, **final approval will be given by the tutor** to ensure balance and fairness across all groups.

When approaching the project, you should follow the **software life cycle**, starting with a user-centered design process. This includes researching the problem, planning features, and creating both low- and high-fidelity prototypes. Your design *may evolve over time*; each group must maintain clear documentation of all design decisions which will be tracked at each stage of the project.

All members of the team must have a strong understanding of the entire application. This means **close collaboration** is essential, as each student should be able to present and explain any part of the app during the final demonstration.

1. **Design Document:** Marked out of 20 according to the Design Document rubric, but weighted to **10%** of the course grade.
2. **Prototype:** Marked out of 20 according to the Project Prototype rubric, but weighted to **10%** of the course grade.

Design Document – marked out of 20; weighted 10% – Learning outcomes assessed: LO1, LO3

This phase focuses on planning and designing your application based on the selected client brief. You will submit a single **PDF document** as a group. The Design Document is **marked out of 20** according to the rubric, but it is **weighted to 10%** of the course grade. It is assessed across five areas: Background Research, Features List, Mockups, Innovation, and Document Organization and Presentation.

a. Background Research (Group Assessment) – 4%

- A clear explanation of the real world or context, including who is affected and why the issue matters.
- A clear description of the app's purpose, goals, and intended impact as a solution to the problem.
- Supporting research or references, such as online research, trends, or statistics.
- Group interview summary with your tutor to validate and refine your understanding, focused on collecting information from your tutor within class time. Include questions asked and summary of responses.
- Explain what gives your app a competitive edge compared to similar apps in the market.

This section sets the foundation for your design choices and should demonstrate your understanding of the user needs and problem domain.

b. Features List – 4%

- **Name of Feature**
- **Main Task:** What is the feature trying to accomplish from the user's perspective?
- **User:** Who is the intended user for this feature, such as regular user, admin, employee, or manager?
- **Sub-Tasks/Steps:** Describe a brief flow or individual steps required to complete the feature.
- **Developer:** The team member responsible for designing and implementing the feature.
- **Tester:** Another team member responsible for testing the feature.

Use a **clear, structured table or bullet-point format** to ensure all features are well defined. Every student must be responsible for at least **two features as a developer**, and the workload should be fairly distributed between team members. The feature list should also include at least one advanced feature agreed upon with the tutor, such as LLM integration, maps, group chats, or real-time multiplayer. The advanced feature should be assigned to one or a maximum of two developers.

c. Mockups – 8%

- Clearly show and reflect the structure and flow of each feature.

- Include multiple screens if the feature requires them.
- Contain **written descriptions** for each screen explaining its purpose and layout.
- Identify and label **all UI elements**, such as buttons, inputs, and labels, and explain their function.

You may use **any software or method** for creating low-fidelity mockups, including paper sketches scanned, **Draw.io**, **Figma**, or **Canva**. The mockups should include all necessary screens, show clear navigation or screen flow, label UI elements clearly, and include written screen descriptions explaining each screen's purpose and layout. Marks are awarded for coverage of assigned features, clarity of screen flow, usability of layout, meaningful UI labelling, and overall neatness/professionalism.

d. Innovation (Group Assessment) – 2%

- Demonstrate creative and thoughtful design ideas beyond the basic requirements.
- Show evidence of original thinking, user-focused improvements, or meaningful enhancement of the selected brief.
- Marks are awarded for relevance, feasibility, and added value to the proposed app solution.

e. Document Organization and Presentation (Group Assessment) – 2%

- The design document should be clearly structured and easy to follow.
- Use consistent formatting throughout the document.
- Present the document professionally with good readability.
- Marks are awarded for organization, consistency, readability, and overall presentation quality.

Project Prototype – marked out of 20; weighted 10% – Learning outcomes assessed: LO1, LO2, LO3

Your team must develop a **high-fidelity interactive prototype** that accurately represents the final design of your application. This prototype should bring your mockups to life with full user flows and realistic screen transitions. You are expected to use **Figma** for this assessment; other tools may be permitted with approval given by the tutor.

The Project Prototype is **marked out of 20** according to the rubric, but it is **weighted to 10%** of the course grade.

- All **main features** of the app are designed and linked based on your feature list.
- **Interactive links** between screens simulate taps, button presses, and navigation.
- A **clearly structured prototype** reflects real-world app behavior and layout.
- **Visual consistency** is maintained in terms of color themes, fonts, and UI elements.
- **Screen organization** is consistent in the prototype, and all frames/screens must be clearly named by developer, such as Login_Ahmed_2022XXXXX.

You are required to submit the **interactive prototype file** using a Figma file (.fig) alongside the shared link. It is encouraged that your prototype is tested thoroughly by team members and real users to gather feedback which ensures that it is up to standard.

Your design should follow key **UI/UX principles** to ensure clarity, usability, and aesthetic appeal. A strong prototype demonstrates not only good visual layout but also realistic interaction behavior and awareness of user needs. Consider both **how things look** and **how users will interact** with the application.

Visual Design Principles

- **Balance** – Use symmetry or asymmetry effectively to achieve visual stability.
- **Contrast** – Ensure that UI elements stand out appropriately and are distinguishable through color, size, and shape.
- **Alignment** – Maintain consistent vertical and horizontal alignment for UI elements to support readability and structure.
- **Simplicity** – Avoid clutter, prioritize clarity and ease of use.
- **Proximity** – Group related UI elements to indicate logical relationships.
- **Repetition** – Maintain consistency in layout, icons, and styles across each screen.
- **White Space** – Use spacing to improve focus, reduce overload, and organize content.

Interaction & UX Design Practices

- **Clear Navigation** – Users should easily understand how to navigate between screens.
- **Interactive Elements** – Buttons, links, and input fields must function as expected.
- **Error & Feedback States** – User feedback, success, and error screens or messages should be included in the prototype.
- **Mobile Optimization** – Ensure interactive areas are suitable for mobile use.
- **Consistency** – Apply the same styles and behaviors across all screens and features.

Project Brief Selection

Each group must choose **one** of the following project briefs as the foundation for their app design. These briefs are designed to reflect **real-world problems** aligned with the Sustainable Development Goals (SDGs), and they offer a balance of design complexity, user roles, and potential technical challenges.

To ensure a diverse set of projects within each class section, a **maximum of two groups per section** may select the same brief. Briefs will be assigned on a **first-come, first-served basis**, but **all selections must be approved by the tutor** before work can begin. Approval ensures fair distribution, feasibility, and alignment with assessment criteria.

Once approved, your group will base all project deliverables, including the design document, mockups, and prototypes on the selected brief. Choose wisely based on your team's interests, skill sets, and ideas for innovation.

Important Note: The client briefs provided are not exhaustive. They are designed to give you a starting point, but **not all features, flows, or requirements are fully specified**. You are expected to conduct **additional research**, explore real-world references, and demonstrate **innovation** in your design. The **interview component** of the project is an opportunity to further clarify needs, identify missing details, and refine your ideas based on feedback.

Project Identity Requirements

- A **unique app name** that is memorable and reflects your solution.
- A **basic logo/icon concept** that can be refined in later phases.
- **Consistent use of this identity** across all project materials, including app name on document covers and headers, logo/icon in mockups and prototype screens, and the same visual identity throughout all three assessment phases.

Project Brief 1: Startup Co-founder & Team Formation Platform

Overview

This project asks students to design a mobile platform that helps founders and potential collaborators discover startup ideas, connect based on skills and interests, and form teams to develop early-stage ventures.

The solution should go beyond traditional networking by supporting idea sharing, team matching, collaboration, mentor feedback, and progress toward launch.

Example User Roles and Main Features

User Roles: Founder, Collaborator, Mentor/Investor, Admin.

Main Features: Startup idea posting, skill-based profile creation, collaborator discovery, team matching, project invitations, mentor feedback, progress tracking, messaging, milestone planning, and admin moderation.

Additional Considerations/Challenges

- How can users be matched based on skills, interests, and goals?
- How can the platform stand out from networking or recruitment apps?
- How can AI or other technologies improve the user experience?

Related Sustainable Development Goals (SDGs)

- **SDG 8:** Decent Work and Economic Growth
- **SDG 9:** Industry, Innovation and Infrastructure
- **SDG 17:** Partnerships for the Goals

Project Brief 2: Intelligent Learning Platform

Overview

This project asks students to design an AI-supported mobile learning platform that helps students organize materials and turn them into useful study resources such as summaries, notes, quizzes, and flashcards.

The solution should promote active learning, personalization, revision planning, and possible collaboration or sharing between students.

Example User Roles and Main Features

User Roles: Student, Tutor/Teacher, Study Group Member, Admin.

Main Features: Material upload, AI-generated summaries, flashcard creation, quiz generation, study plan scheduling, progress tracking, shared study folders, group revision spaces, resource bookmarking, and admin content management.

Additional Considerations/Challenges

- How can uploaded materials become accurate study resources?
- How can generated resources support different learning styles?
- How can students organize and revisit resources throughout the term?

Related Sustainable Development Goals (SDGs)

- **SDG 4:** Quality Education
- **SDG 9:** Industry, Innovation and Infrastructure
- **SDG 10:** Reduced Inequalities

Project Brief 3: Hobbies & Life Experiences Platform

Overview

This project asks students to design a mobile platform that helps users discover hobbies, activities, events, and experiences in Bahrain while recording and celebrating what they have completed.

The solution should encourage exploration, achievement, social interaction, and meaningful support for individuals and activity providers.

Example User Roles and Main Features

User Roles: Explorer/User, Activity Provider, Community Host, Admin.

Main Features: Activity discovery, location-based recommendations, experience booking, personal hobby log, achievement badges, reviews and ratings, social sharing, provider listings, event calendars, and admin approval tools.

Additional Considerations/Challenges

- How can users be encouraged to keep exploring new hobbies?
- How can social features and achievements motivate participation?
- How can location services or AI improve recommendations?

Related Sustainable Development Goals (SDGs)

- **SDG 4:** Quality Education
- **SDG 8:** Decent Work and Economic Growth
- **SDG 9:** Industry, Innovation and Infrastructure

Project Brief 4: Social Fitness & Wellness Platform

Overview

This project asks students to design a mobile platform that turns fitness and wellness into a shared social experience, helping users stay active, build healthy habits, and support one another.

The solution should combine activity tracking, wellness goals, nutrition, social motivation, and personalized guidance in an engaging way.

Example User Roles and Main Features

User Roles: Member/User, Fitness Coach, Wellness/Nutrition Advisor, Admin.

Main Features: Workout tracking, wellness goals, meal and nutrition logging, friend challenges, community groups, progress sharing, achievement badges, personalized recommendations, coach feedback, and admin moderation.

Additional Considerations/Challenges

- How can activity, nutrition, and wellness be combined clearly?

- How can social motivation keep users active?
- How can AI provide useful wellness guidance?

Related Sustainable Development Goals (SDGs)

- **SDG 3:** Good Health and Well-being
- **SDG 10:** Reduced Inequalities
- **SDG 11:** Sustainable Cities and Communities

Project Brief 5: Social Gaming & Entertainment Platform

Overview

This project asks students to design a mobile social gaming platform that brings together casual multiplayer games, communication, events, and community features in one experience.

The solution should encourage users to discover games, build friendships, join competitions, earn rewards, and return through meaningful social interaction.

Example User Roles and Main Features

User Roles: Player, Game Host, Community Moderator, Admin.

Main Features: Multiplayer game lobby, friend invitations, in-game chat, tournament creation, leaderboards, seasonal events, player profiles, achievement rewards, community spaces, and moderation tools.

Additional Considerations/Challenges

- How can rewards and progression encourage long-term play?
- How can users discover suitable games, groups, and players?
- How can AI personalize play, recommendations, or moderation?

Related Sustainable Development Goals (SDGs)

- **SDG 3:** Good Health and Well-being
- **SDG 9:** Industry, Innovation and Infrastructure
- **SDG 10:** Reduced Inequalities

Project Brief 6: Multiplayer Knowledge & Challenge Platform

Overview

This project asks students to design a mobile platform for multiplayer knowledge-based games and challenges that test reasoning, memory, creativity, language, and general knowledge.

The solution should support varied challenge types, competition, progression, rankings, tournaments, and fresh content that keeps users engaged over time.

Example User Roles and Main Features

User Roles: Player, Quiz Creator, Challenge Host, Admin.

Main Features: Multiplayer quiz rooms, challenge categories, timed competitions, AI-generated questions, custom quiz creation, rankings, tournaments, achievement badges, progress history, and admin question review.

Additional Considerations/Challenges

- How can rewards and progression encourage improvement?
- How can varied challenges suit different interests and skill levels?
- How can AI generate fresh content or intelligent opponents?

Related Sustainable Development Goals (SDGs)

- **SDG 4:** Quality Education
- **SDG 9:** Industry, Innovation and Infrastructure
- **SDG 10:** Reduced Inequalities

Project Deliverables

Phase 1: Design Document

Section	Marking Requirements	Marks
Background Research	- Explain the real-world problem, target users, and why it matters. - State the app purpose, goals, and intended impact. - Use relevant research or references to support the idea. - Identify the app's competitive edge.	4
Features List	- Submit a structured feature list. - Include feature name, user role, main task, flow, developer, and tester. - Use at least three user roles. - Each student must develop at least two features. - Include one approved advanced feature assigned to one or two developers.	4
Mockups	- Submit low-fidelity mockups for assigned features. - Include all required screens and clear navigation flow. - Label key UI elements clearly. - Add brief screen descriptions explaining purpose and layout. - Marks focus on coverage, usability, clarity, labelling, and neatness.	8
Innovation	- Show creative ideas beyond the basic requirements. - Demonstrate original, user-focused improvements. - Marks focus on relevance, feasibility, and added value.	2
Document Organization and Presentation	- Organize the document clearly and logically. - Use consistent formatting and professional presentations.	2

	Ensure readability, neatness, and overall quality.	
Total		20

Note: Although the Design Document is marked out of 20, the final contribution to the course grade is weighted to **10%**.

Phase 2: Prototype

Section	What is required to be submitted	Marks
Prototyping	Submit a high-fidelity interactive prototype using Figma . Include links between screens simulating user interaction. Submit both the interactive file (.fig) and the shared URL in a text document.	20
Total		20

Note: Although the Prototype is marked out of 20, the final contribution to the course grade is weighted to **10%**.